PRIYASH BARYA

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EDUCATION

University of Illinois at Urbana-Champaign Champaign, USA
Ph.D. Candidate at Electrical and Computer Engineering (*GPA* - 4/4) Aug 2020 - Present
Proposed Thesis title - Ultra slow light in nanophotonics
Advisor - Prof. Elizabeth A. Goldschmidt

MS Electrical and Computer Engineering (*GPA* - 4/4) Aug 2020 - Dec 2022
Thesis title - Photonic-plasmonic coupling enhanced fluorescence for digital-resolution protein detection
Advisor - Prof. Brian T. Cunningham

Birla Institute of Technology & Science - Pilani Hyderabad, India
B.E. Electronics and Communication Engineering (*GPA* - 9.5/10) Aug 2016 - July 2020
& Minor in Physics
Thesis title - Aluminum Doped ZnO Nanodisk and Silver Dimer Plasmonics
Advisor - Prof. Ambarish Ghosh

RESEARCH EXPERIENCE

Interests : Quantum memory, Quantum networks, Materials for Quantum, Quantum nanophotonics

University of Illinois at Urbana-Champaign Champaign, USA
Joan and Lalit Bahl Fellow, Graduate Research Assistant, Physics Department Aug 2022 - Present
Advisor - Prof. Elizabeth A. Goldschmidt

- Achieved ultra-high optical Q ($\sim 10^8$) tunable resonances in Er^{3+}:LNOI microrings via slow-light–enhanced spectral-hole burning, a 3-order of magnitude increase in Q.

- Experimentally discovered and theoretically modeled an intensity-dependent dephasing mechanism, driving Er^{3+} toward the radiative limit ($T_2 \rightarrow 2T_1$) by decoupling it from the spin bath.

- Developed telecom quantum memory, with 1.1 μs storage, and upto 2.9% efficiency, and 2.2 GHz bandwidth, and noise below a single photon, record high efficiency-storage time product on any integrated nanophotonic platform.

- Discovered superradiant and other collective effects in waveguide coupled ensemble of disordered solid state emitters (Manuscript in prep)

- Collaborating with Prof. Mohammad Hafezi's group on topological poling-free nested frequency-phase matching for TFLN integrated nonlinear photonics (Manuscript in prep)

University of Illinois at Urbana-Champaign Champaign, USA
Graduate Research Assistant, Electrical and Computer Engineering Department Aug 2020 - Aug 2022
Advisor - Prof. Brian T. Cunningham

- Developed and characterized plasmonic–photonic hybrid experiments that achieved 52× fluorescence enhancement and enabled single protein detection of IL-6 molecules down to $\sim$ 10 fg/mL, 3-orders lower than conventional ELISA.

- Performed electromagnetic simulations validating photonic-crystal–enhanced single-quantum-dot fluorescence for 10 aM biomarker detection of target DNA.

Indian Institute of Science Bengaluru, India
Project Assistant, Centre for Nano Science and Engineering Dec 2019 - July 2020

Advisor - Prof. Ambarish Ghosh

Optimized and fabricated AZO plasmonic nanostructures via RF-sputtering and nanosphere lithography, and identified Fano-resonant hybrid modes in silver–graphene dimers using COMSOL.

JOURNAL PAPERS

Google Scholar (Citations : 370, *h*-index : 7, as of Aug 2026)

1. **P. Barya***, D. Chen*, A. Prabhu, L. Heller, E. Chow, H. Kim, J. Akin, V. Niaouris, J. Zhang, A. Dibos, P. Wang, E.A. Goldschmidt "Telecom quantum memory over one microsecond in nanophotonic lithium niobate", Accepted in **Nano Letters** (2026) , arxiv Link

2. **P. Barya**, A. Prabhu, L. Heller, E. Chow, E.A. Goldschmidt "Ultra High-Q tunable microring resonators enabled by slow light", **Nature Communications** 16, 10496 (2025) DOI - 10.1038/s41467-025-65533-1

   Highlighted by **News** - Phys.org, EurekaAlert, MSN, Mirage News, Semiconductor Engineering, IQUIST News, UIUC Physics News, Photonics Online

3. **P. Barya***, Y. Xiong*, S. Shepherd*, R. Gupta, L. Akin, J. Tibbs, H. Lee, S. Singamaneni, B. Cunningham "Photonic-Plasmonic Coupling Enhanced Fluorescence Enabling Digital-Resolution Ultrasensitive Protein Detection", **Small** (2023): 2207239, DOI - 10.1002/smll.202207239

   Selected as **Frontpiece Cover article**, Highlighted by **News** - Cancer Center at Illinois, ECE News, IGB-UIUC, HMNTL, Bioengineering News

4. Y. Xiong , Q. Huang , T. Canady , **P. Barya** , O. Arogundade , C. Race , S. Liu , A. Smith , M. Kohli, B. Cunningham, "Photonic crystal enhanced fluorescence emission and blinking suppression for single quantum dot digital resolution biosensing", **Nature Communication** 13(1), 4647, 2022, DOI - 10.1038/s41467-022-32387-w

5. N. Li, X. Wang, J. Tibbs, C. Che, A. Peinetti, B. Zhao, L. Liu, **P. Barya**, L. Cooper, L. Rong, X. Wang, Y. Lu, B. Cunningham, "Label-Free Digital Detection of Intact Virions by Enhanced Scattering Microscopy", **Journal of the American Chemical Society (JACS)**, 144(4), 1498-1502, 2021, DOI - 10.1021/jacs.1c09579

6. Y. Xiong, N. Li, C. Che, W. Wang, **P. Barya**, W. Liu, L. Liu, X. Wang, S. Wu, H. Hu and B. Cunningham "Microscopies Enabled by Photonic Metamaterials", **Sensors**, 22, no. 3 (2022): 1086, DOI - 10.3390/s22031086

7. A. P. Manuel, **P. Barya**, S. Riddell, S. Zeng, K. Shankar, "Plasmonic photocatalysis and SERS sensing using ellipsometrically modeled Ag nanoisland substrates", IOP **Nanotechnology**, 31, no. 36 (2020): 365301, DOI - 10.1088/1361-6528/ab814c

8. V. Selamneni, **P. Barya**, N. Deshpandey, P. Sahatiya, "Low-cost, disposable, flexible and smartphone enabled all paper based pressure sensor for monitoring drug dosage in smart medicine applications", **IEEE Sensors Journal**, 19, no. 23 (2019): 11255-11261, DOI - 10.1109/JSEN.2019.2935383

SELECTED CONFERENCES/TALKS

1. **P. Barya**, D. Chen, A. Prabhu, L. Heller, E. Chow, H. Kim, J. Akin, V. Niaouris, J. Zhang, A. Dibos, P. Wang, E.A. Goldschmidt "Integrated Nanophotonic Waveguides for Telecom-band Single Photon Storage " , **CLEO** 2026

2. **P. Barya**, A. Prabhu, Laura Heller, E. Chow, E. A. Goldschmidt, ‘Ultra high-Q tunable microring resonators enabled by slow light ”, presented in **SPIE Photonics West** 2026 [**Invited Presentation**], **SPIE Optics and Photonics** 2025, **CLEO** 2025, IQUIST Young Researchers Seminar, University of Illinois
3. **P. Barya**, A. Prabhu, O. Orsel, E. Chow, E. A. Goldschmidt, “Towards a thin-film lithium niobate nanophotonic cavity-enhanced telecom-compatible quantum memory” **SPIE Optics and Photonics** 2024 [**Invited Presentation**], **CLEO** 2024
4. Y. Xiong*, **P. Barya***, S. Shepherd*, R. Gupta, L. Akin, J. Tibbs, H. Lee, S. Singamaneni, B. Cunningham,“Plasmonic-Photonic Hybrid System for Enhanced Fluorescent-based Digital Resolution Biosensing”, **International Conference on Surface Plasmon Photonics**, SPP10 2023
5. **P. Barya**, T. Srinivas, R. Prajesh, “Predictive Modeling of silicon nanowire based sensing for cTroponin-I detection” (Proceedings of International Conference on Computing for Sustainable Global Development, **INDIACom**, 2019) [Link]

PATENTS

1. “Ultra-narrow tunable resonators on-chip enabled by slow light”, **P. Barya**, A. Prabhu, L. Heller, E. Goldschmidt, U.S. Patent Application No.: 63/893,290, *pending* (2025)

FELLOWSHIP AND AWARDS

Joan and Lalit Bahl Fellowship — 2025-2027
Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign

SPIE Optics and Photonics Scholarship — 2026
SPIE

Mavis Future Faculty Fellow — 2025-2026
The Grainger College of Engineering , University of Illinois at Urbana-Champaign

Paul D. Coleman Outstanding Research Award — 2025 & 2026
Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign

Mitacs Globalink Research Scholar — 2019
Mitacs, Canada

National Talent Search Examination Scholarship — 2014-2020
National Council of Educational Research and Training, India

Merit Scholarship — 2016-2020
Birla Institute of Technology & Science - Pilani

TEACHING POSITIONS AND MENTORSHIP

Graduate Teaching Assistant — University of Illinois
Course - ECE 110 - Introduction to Electronics

Graduate Teaching Assistant — BITS Pilani
Course - Electromagnetic Theory

Goldschmidt Lab Mentorship — University of Illinois
3 graduate students and 3 undergraduate students

OUTREACH

- Actively involved in outreach events by IQUIST, HQAN, CQE at UIUC to showcase quantum phenomena to the general public.
- Volunteered in the IISc open house 2020 to showcase science and engineering projects to the public
- Volunteered in the RF/ Microwave Active Circuit Design Workshop 2019 at BITS Pilani, Hyderabad.
- Cognitio (2017) - visiting numerous schools in Bangalore to assist students with career counseling

LEADERSHIP

– **Head of The Movie Club BITS Pilani-** Manager of the Catharsis Film Festival which had a 200+ pan India participation. The Club of over **50 members** also carries out short film making and film discussions. Made **4 Short Films** (Links) - Hyderabad Blues, Imagine, Furore, Epiphany

SERVICE

Peer Reviewer - Nature Physics, Physical Review Letters, npj Nanophotonics, npj Quantum Information, Nano Letters, AIP Advances, The Japan Society of Applied Physics